DA2402 DATA CURATION AND VISUALISATION · IIT MADRAS · 2026

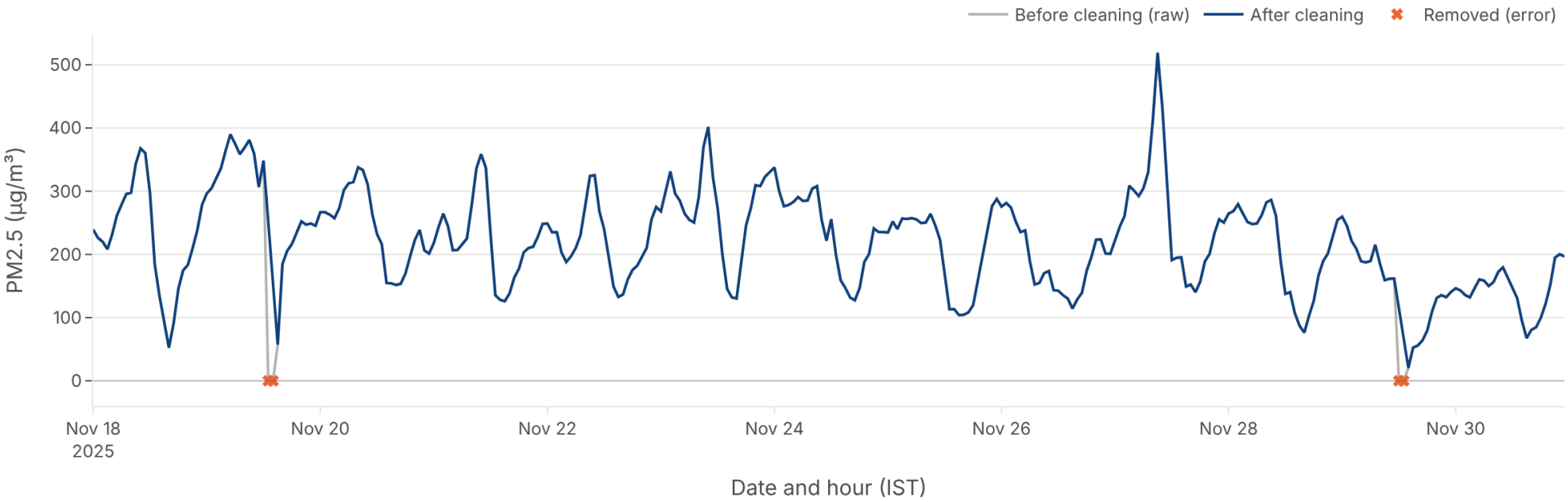
Diwali, stubble burning and Delhi's air

How much of the October–November PM2.5 rise in Delhi comes from Diwali night, and how much from the season's weather?

All 21 charts from our dashboard, in the same order, each with the note that sits under it on the website.
Interactive version: riyan-aggarwal.github.io/Delhi_pollution_level

Riyan Aggarwal	Data visualisation
Shaurya Gupta	Data collection
Prakhar Srivastava	Data cleaning

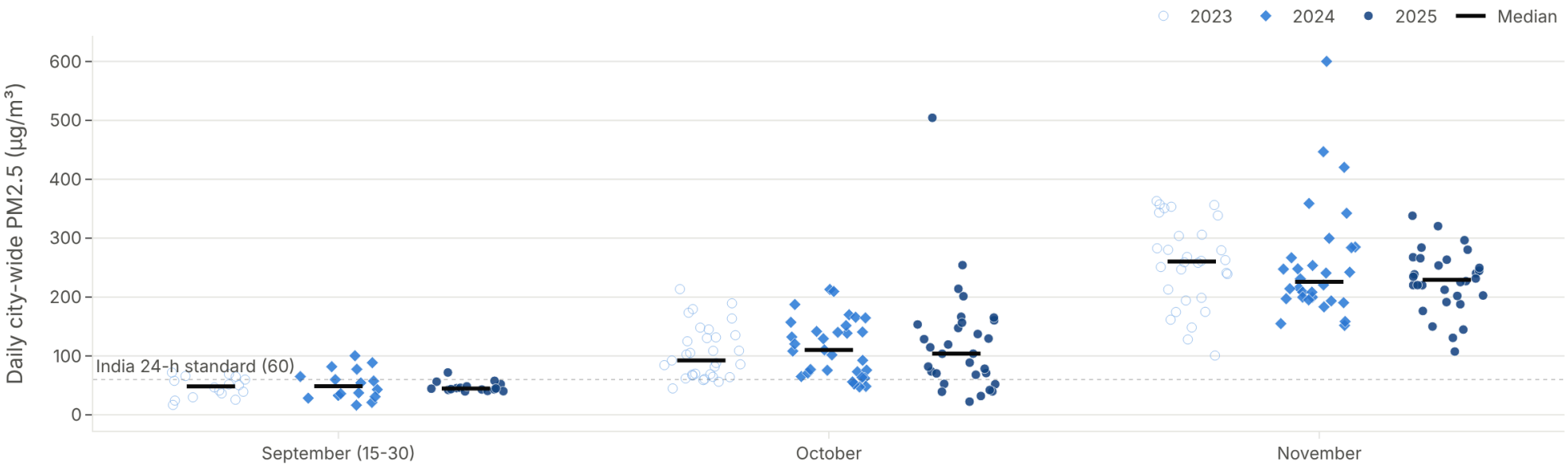
Cleaning removed impossible readings: PM2.5 of exactly zero at ITO, late November 2025



Before and after cleaning. At ITO in late November 2025 the raw data has runs of PM2.5 = 0.00 in the middle of a polluted week, which is impossible. Those hours were removed; the rest of the series is unchanged.

FIGURE 2 · THE SEASON

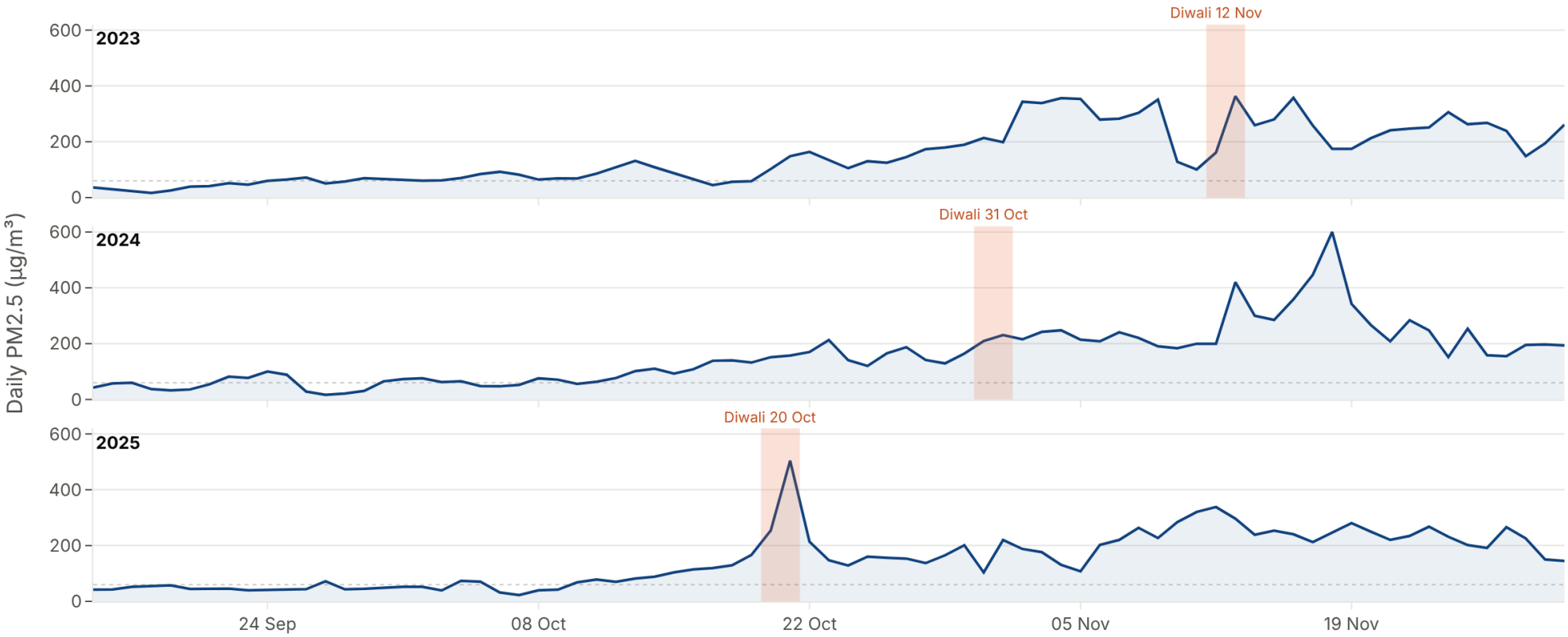
Every day of late September sits near the safe limit; almost every November day is several times above it



Each dot is one day's city-wide average. The medians (black ticks) climb from **45–49 µg/m³ in late September** to **92–110 in October** and **226–260 in November**. Even the cleanest November days are about twice India's limit of 60.

FIGURE 3 · THE SEASON

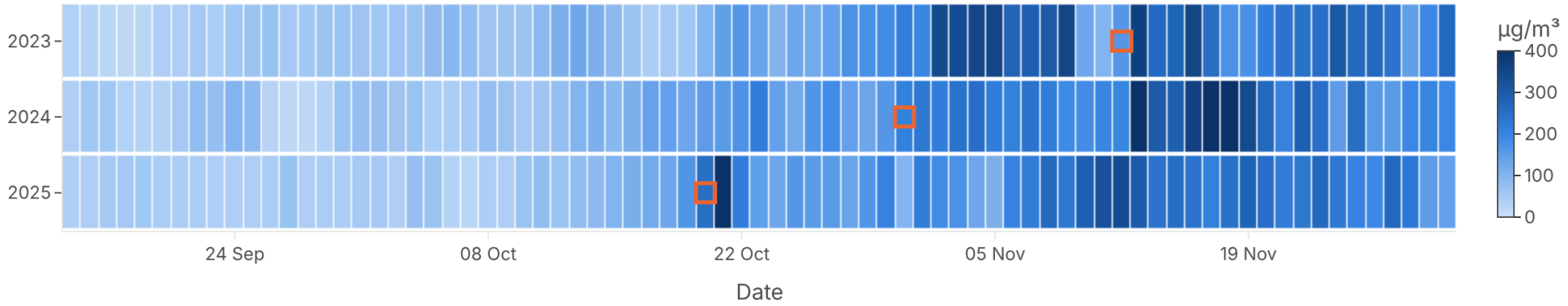
The same climb every year from mid-October; Diwali (orange) lands at a different point on it each time



One panel per year on the same axes (small multiples). The climb starts in mid-October each year whatever the date of Diwali (orange band). The worst day of 2024 was **18 November (600 µg/m³)**, more than two weeks after Diwali; 2025's tallest spike (504) is the day after Diwali.

FIGURE 4 · THE SEASON

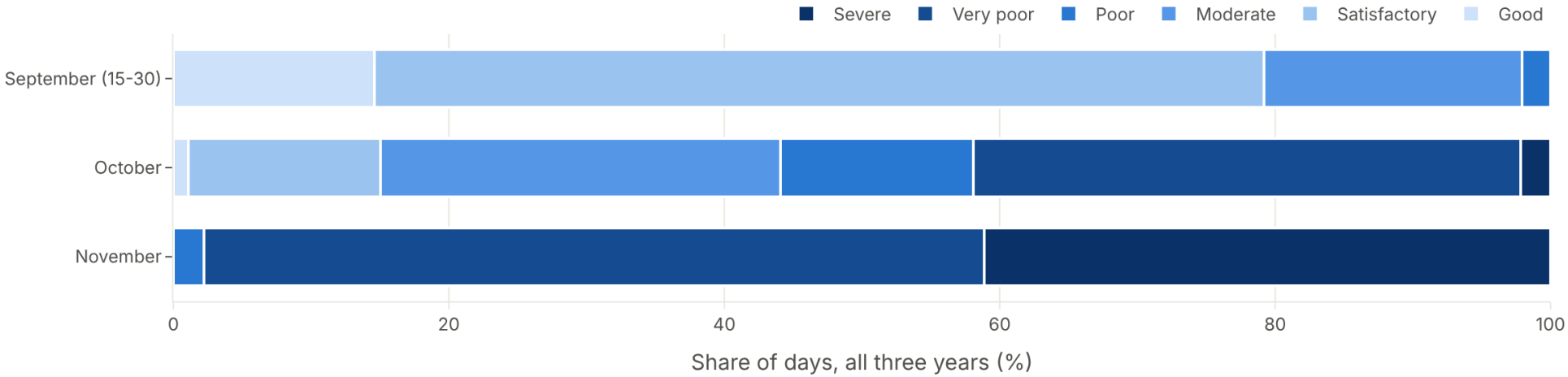
Every season turns dark from mid-October; the first 'very poor' day came on 13, 17 and 18 October



Every day as a cell; darker is dirtier. The first day above 120 µg/m³ ("very poor") came on **13 October 2023, 17 October 2024 and 18 October 2025**.

FIGURE 5 · THE SEASON

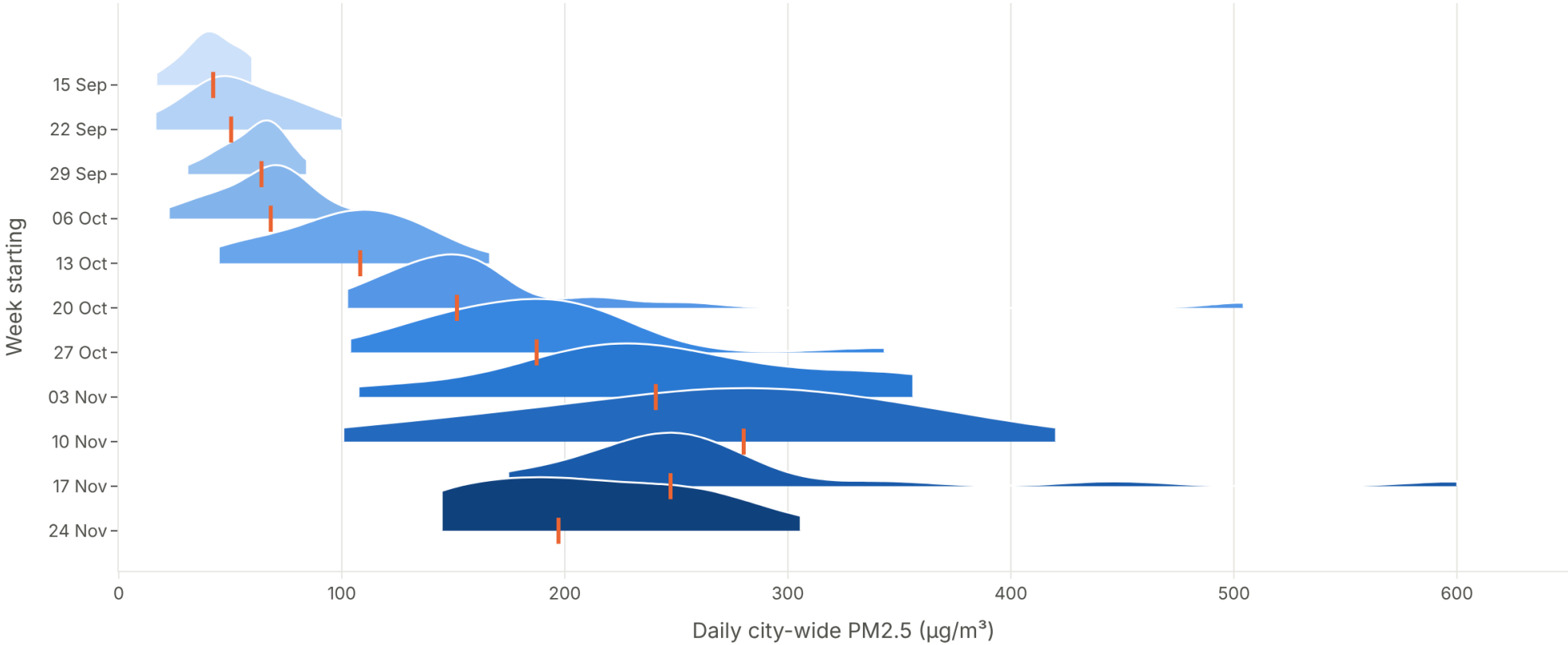
In November, 98% of days were 'very poor' or 'severe'; in late September, none were



India's official AQI bands. In late September **80%** of days were "good" or "satisfactory"; in November **98%** were "very poor" (57%) or "severe" (41%).

FIGURE 6 · THE SEASON

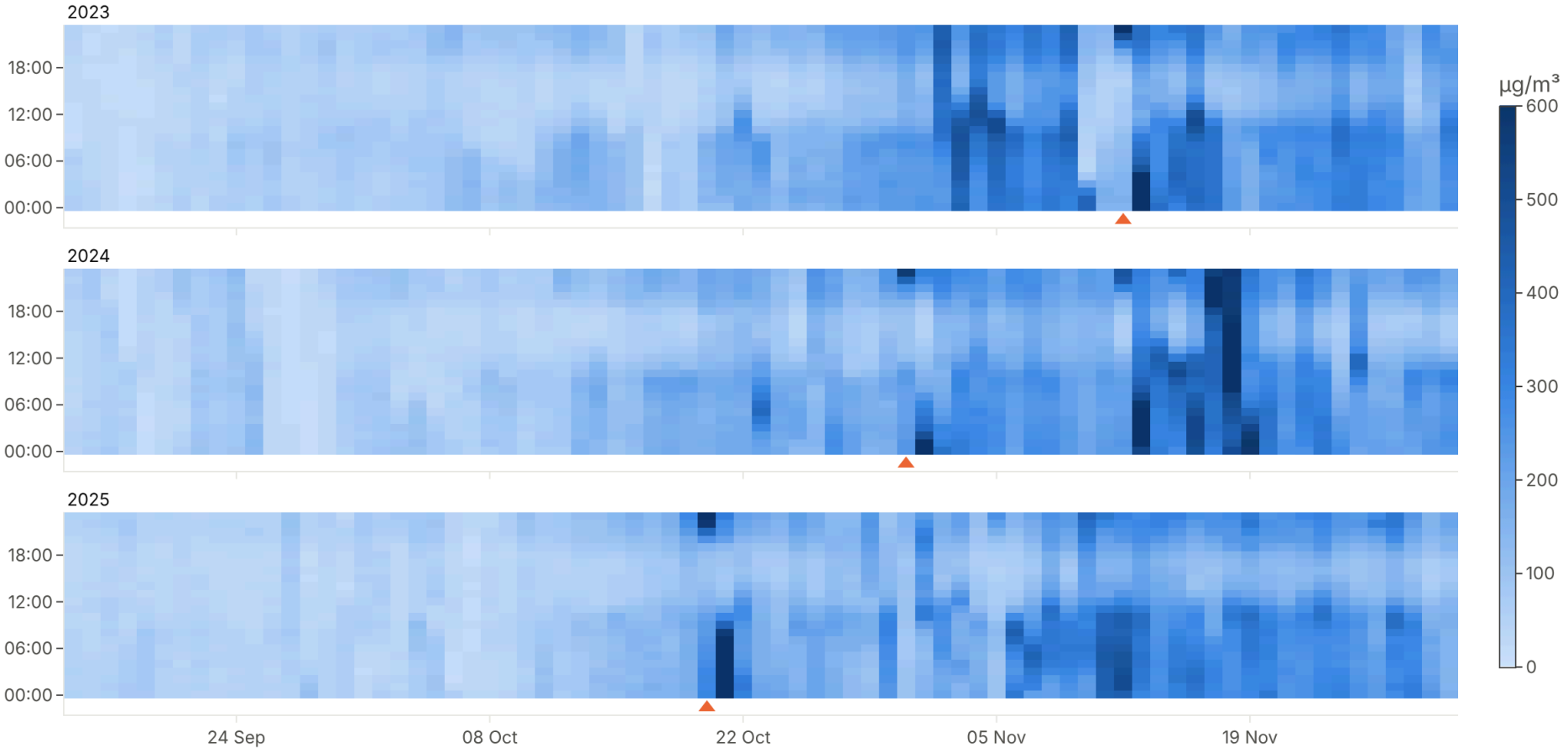
Week by week the whole distribution slides right and spreads out; orange ticks mark each week's median



The spread of daily PM2.5 week by week, all three years together. The whole distribution moves right and widens: the weekly median (orange) goes from **42 µg/m³** in the week of 15 September to **280** in the week of 10 November, then eases to 197 by the end of the month. The thin tails to the right are single extreme days, such as the day after Diwali 2025 and 18 November 2024.

FIGURE 7 · DAY AND NIGHT

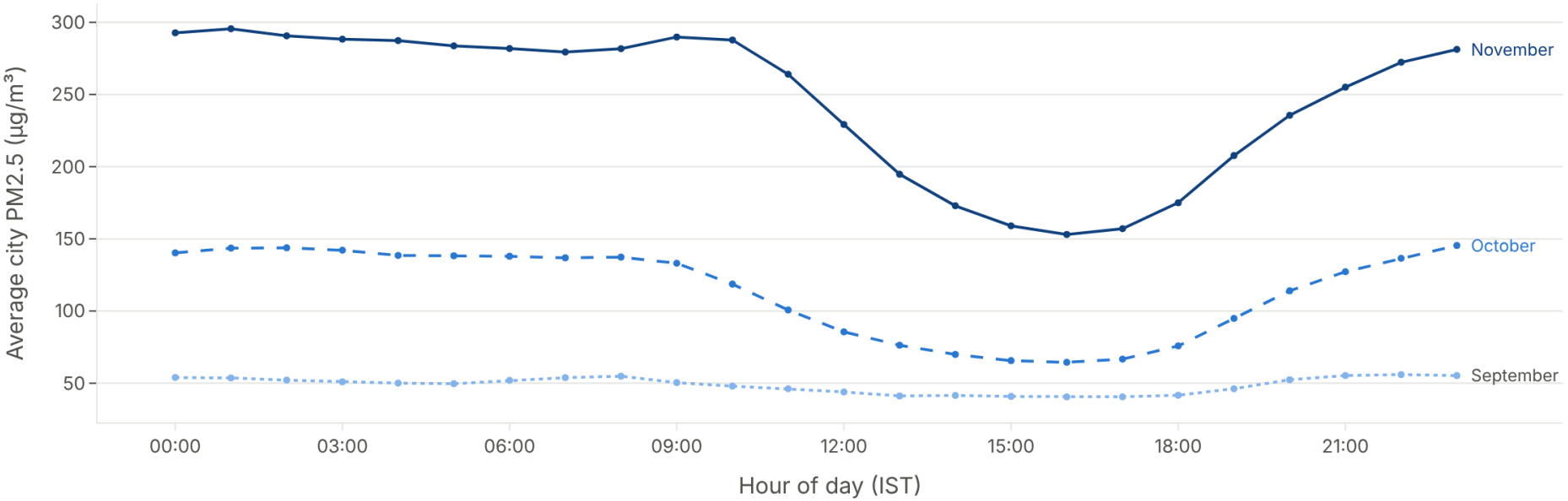
Every night of the season is a dark band; Diwali (orange triangle) is the darkest single column in 2025



Every hour of every season: each column is a day, each row an hour. From late October the night hours turn dark almost every day while the afternoons stay lighter. The darkest single columns are Diwali 2025 (20–21 Oct) and the smog episode of 13–18 November 2024.

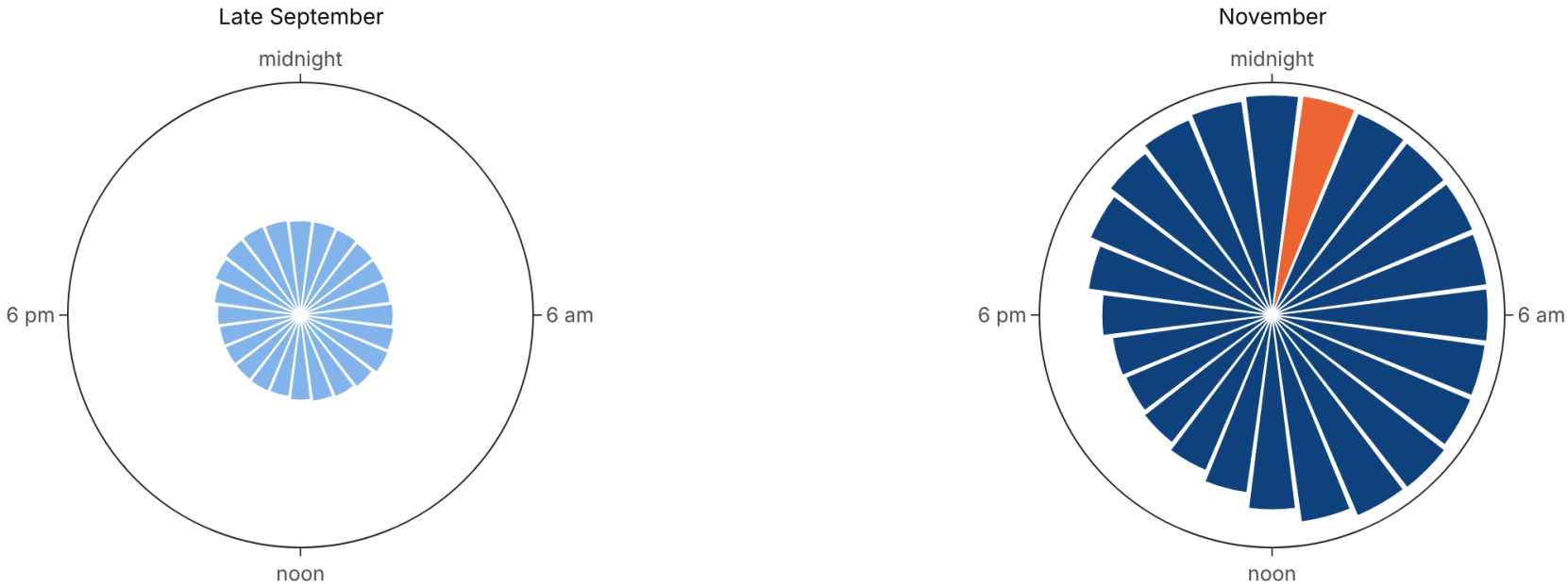
FIGURE 8 · DAY AND NIGHT

Pollution builds through the night and clears in the afternoon; in November the night peak is about twice the afternoon low



Average by hour of day. In November the city is dirtiest around **1 am (296 µg/m³)** and cleanest around **4 pm (153)**, when the sun warms the ground and the air mixes upward. In late September the curve is almost flat, around 40–55.

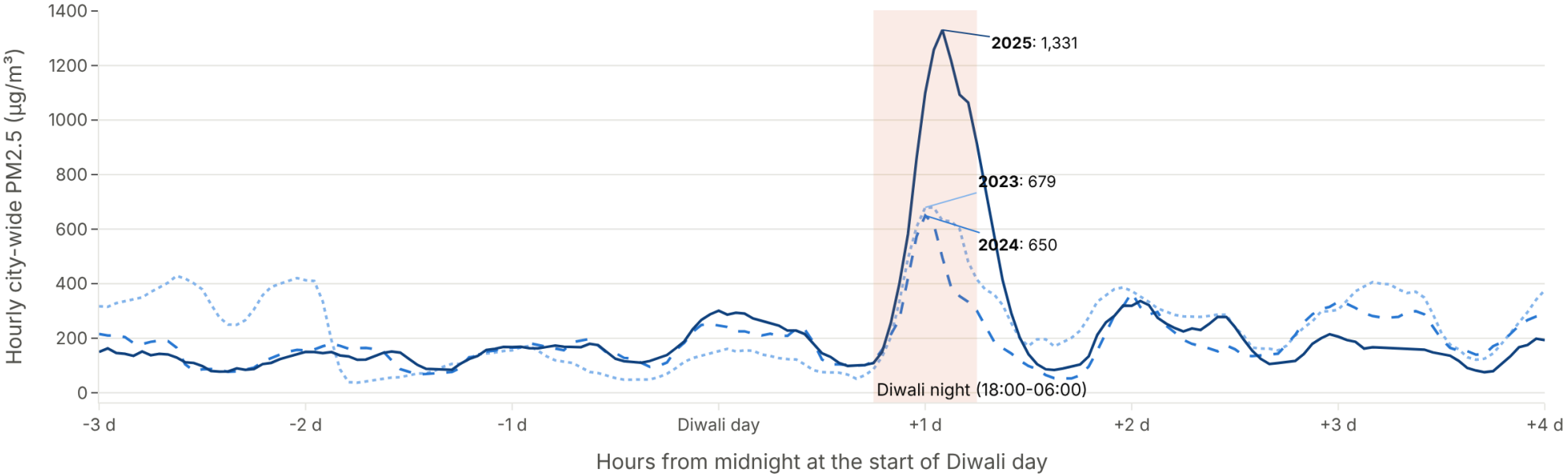
November's air is dirtiest around 1 am (296 µg/m³) and cleanest around 4 pm; late September is small at every hour



The same averages as a 24-hour rose, after Florence Nightingale's 1858 diagram: each wedge is an hour, and its **area** (not its length) shows PM2.5. November's rose dwarfs September's at every hour; the orange wedge is the 1 am peak.

FIGURE 10 · DIWALI NIGHT

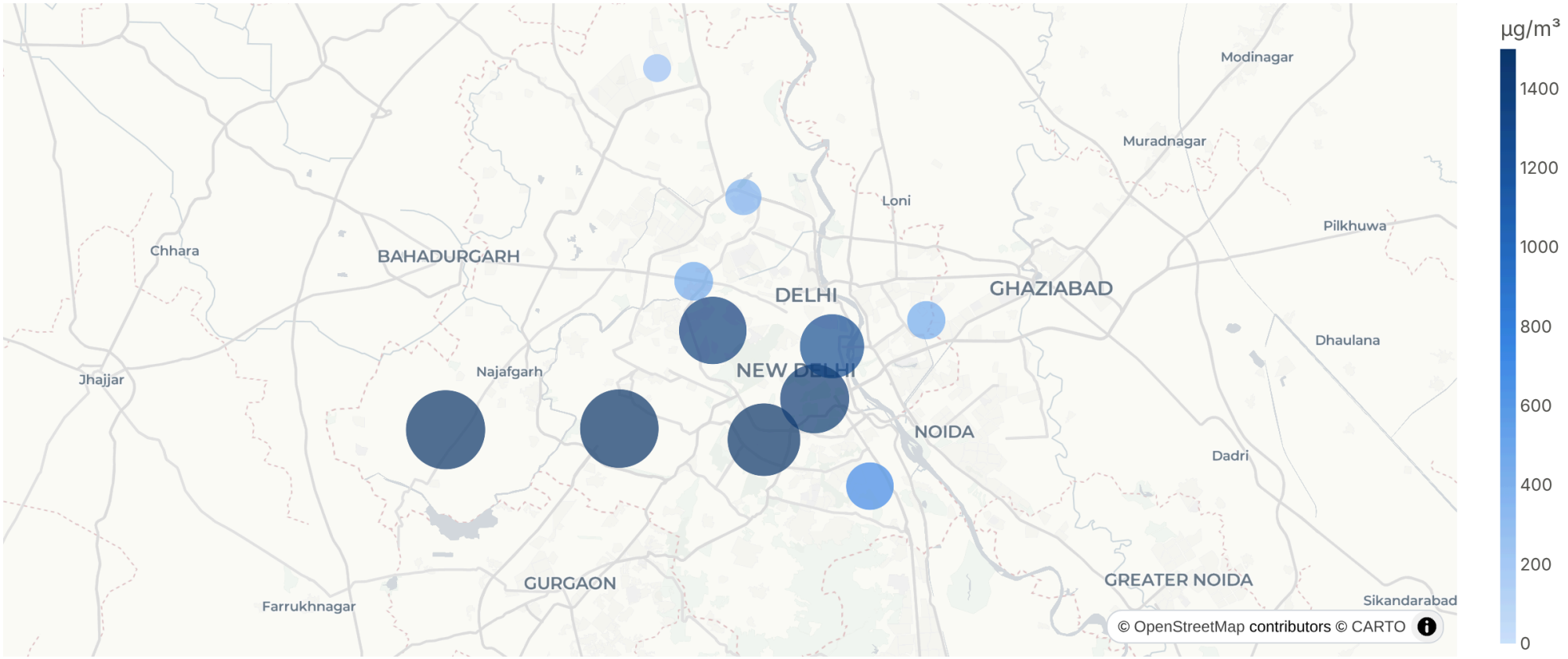
On Diwali night the city's PM2.5 jumps within a few hours, then takes about a day to fall back



City-wide PM2.5 peaked at **679 (2023)**, **650 (2024)** and **1,331 µg/m³ (2025)** between midnight and 2 am, then fell back below 200 by late morning.

FIGURE 11 · DIWALI NIGHT

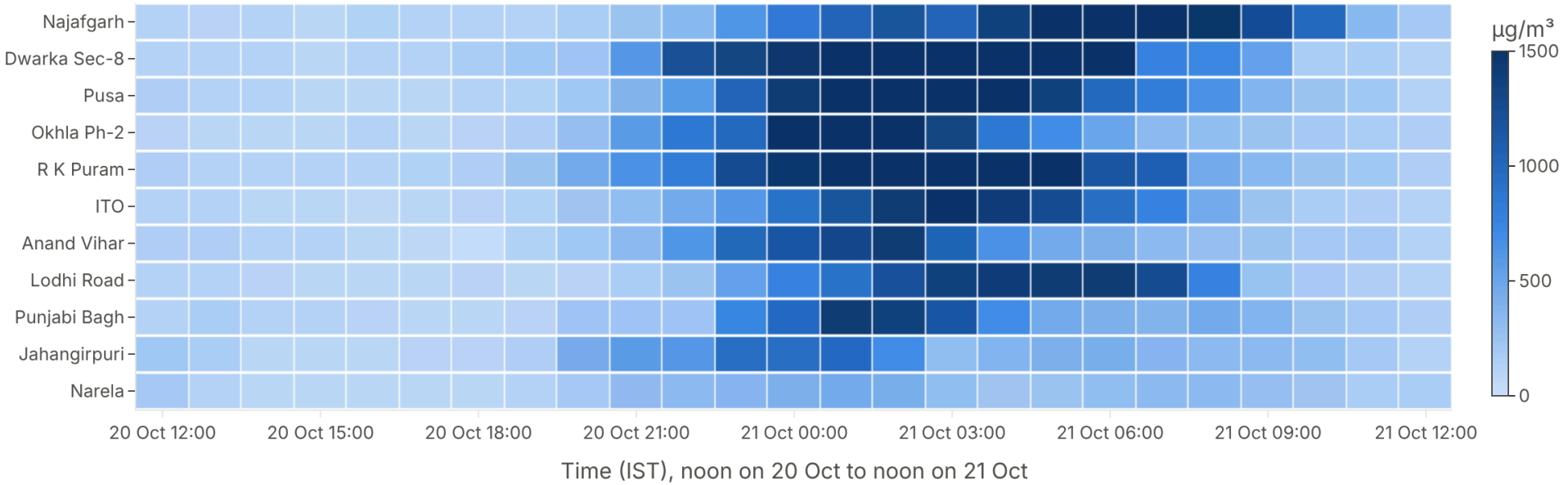
Diwali night 2025: smoke spreads across the city from 9 pm and peaks between 1 and 5 am (shown: 21 Oct 05:00)



Diwali night 2025, shown at 21 Oct 05:00 (on the website this chart is animated hour by hour), in the spirit of John Snow's cholera map and Hans Rosling's moving bubbles. Bubble size and colour show PM2.5. Smoke builds across the city from about 9 pm, and every station peaks between 1 and 5 am; Najafgarh reached **1,901 µg/m³** at 5 am.

FIGURE 12 · DIWALI NIGHT

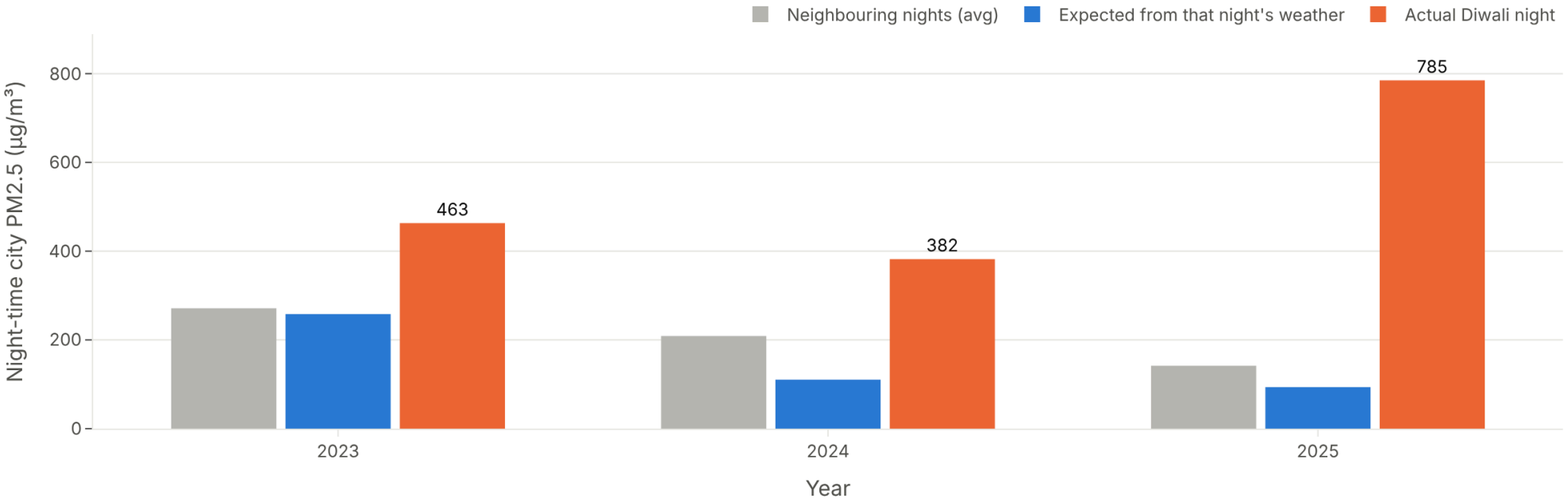
On Diwali night 2025 every station peaked between 1 am and 5 am



The same night as a station-by-hour grid. The core stations turn dark first; Najafgarh, in the south-west, stays dark longest; Narela, on the northern edge, stays far lighter than the rest.

FIGURE 13 · DIWALI NIGHT

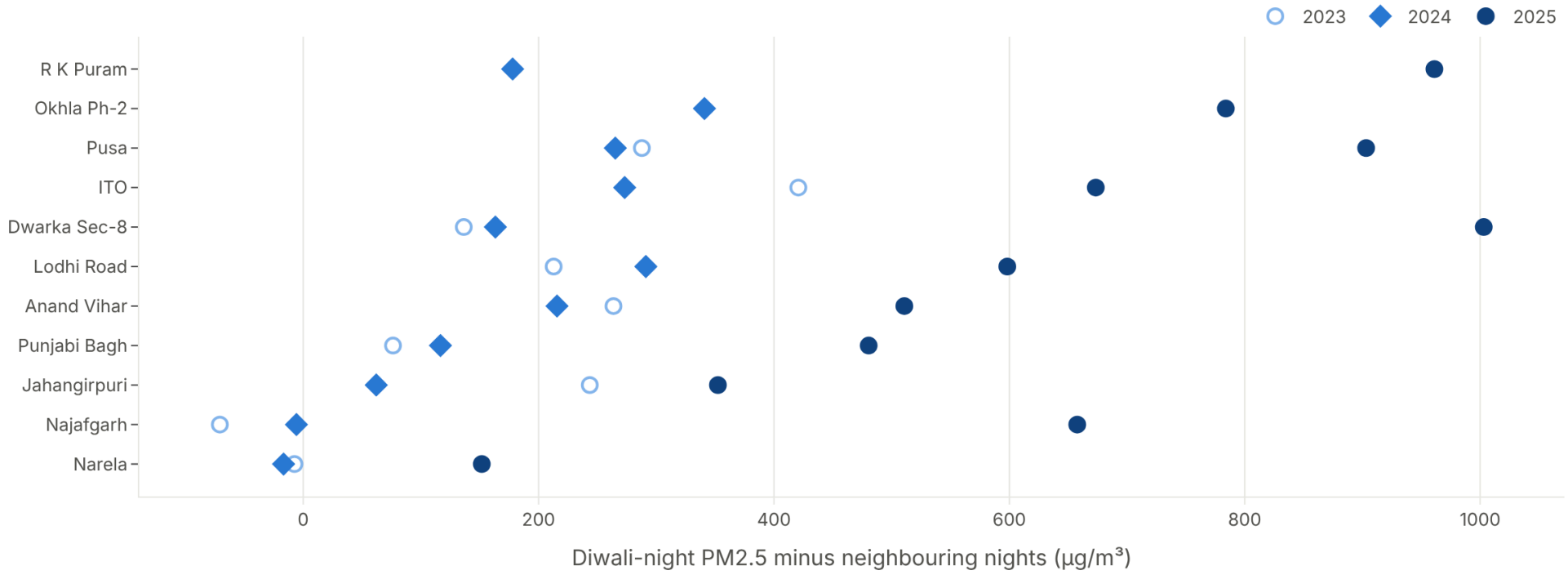
Diwali night was roughly 2 to 8 times what that night's weather would predict



Diwali night averaged **463, 382 and 785 µg/m³**, against 271, 209 and 142 on the surrounding nights. In 2025 it was **8.4 times** what that night's weather predicted; in 2023, when the air was already very still, 1.8 times.

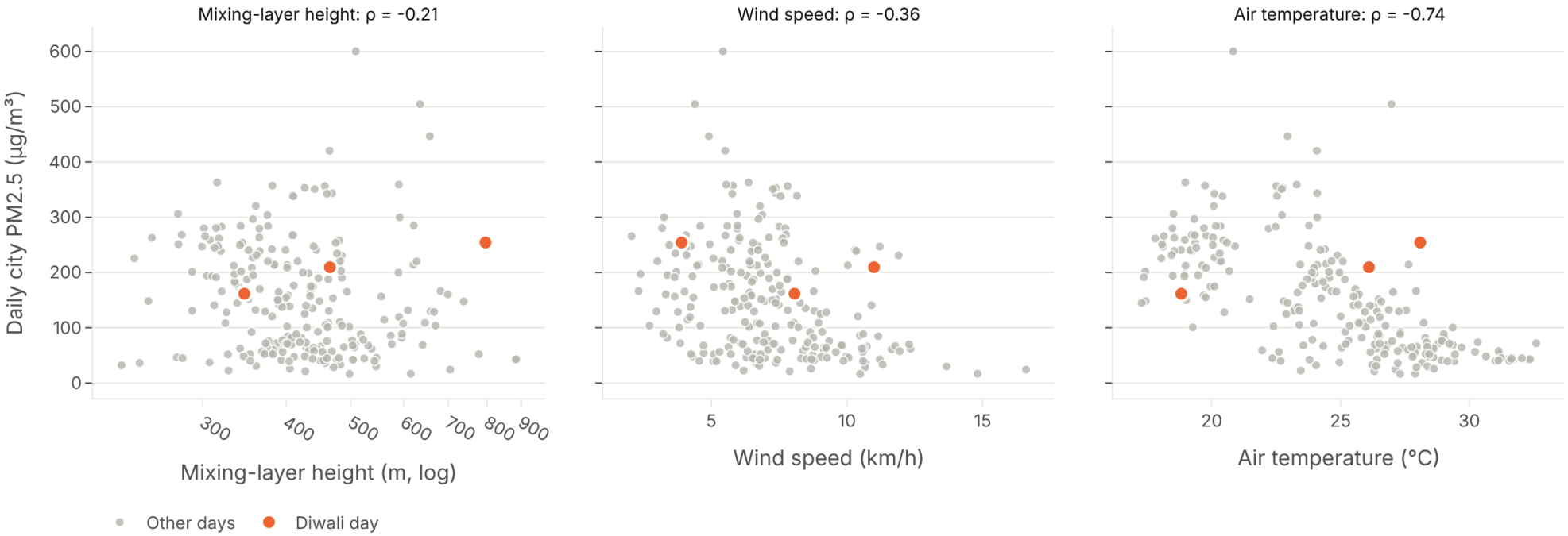
FIGURE 14 · DIWALI NIGHT

2025 was the worst Diwali at almost every station; the outer-edge stations Narela and Najafgarh barely rose in 2023-24



The Diwali-night jump over neighbouring nights ranged from **−71 to +421 µg/m³ in 2023** and **+152 to +1,003 in 2025**. Narela and Najafgarh, on the city's edge, barely rose in 2023 and 2024.

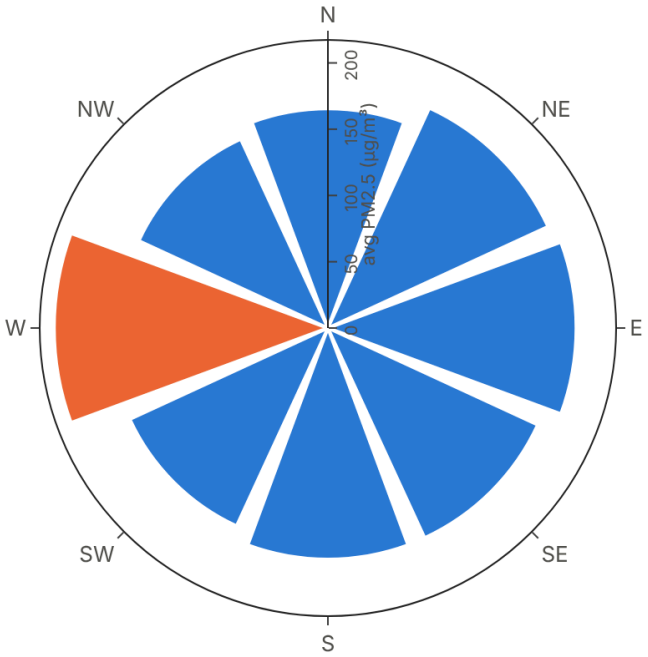
Daily PM2.5 tracks temperature closely; mixing height and wind only loosely



On ordinary days PM2.5 tracks **temperature (ρ = −0.74)** closely, and wind (−0.36) and mixing-layer height (−0.21) only loosely. Temperature also falls steadily through the season while stubble burning and heating rise, so part of this link is "time of year".

FIGURE 16 · WEATHER

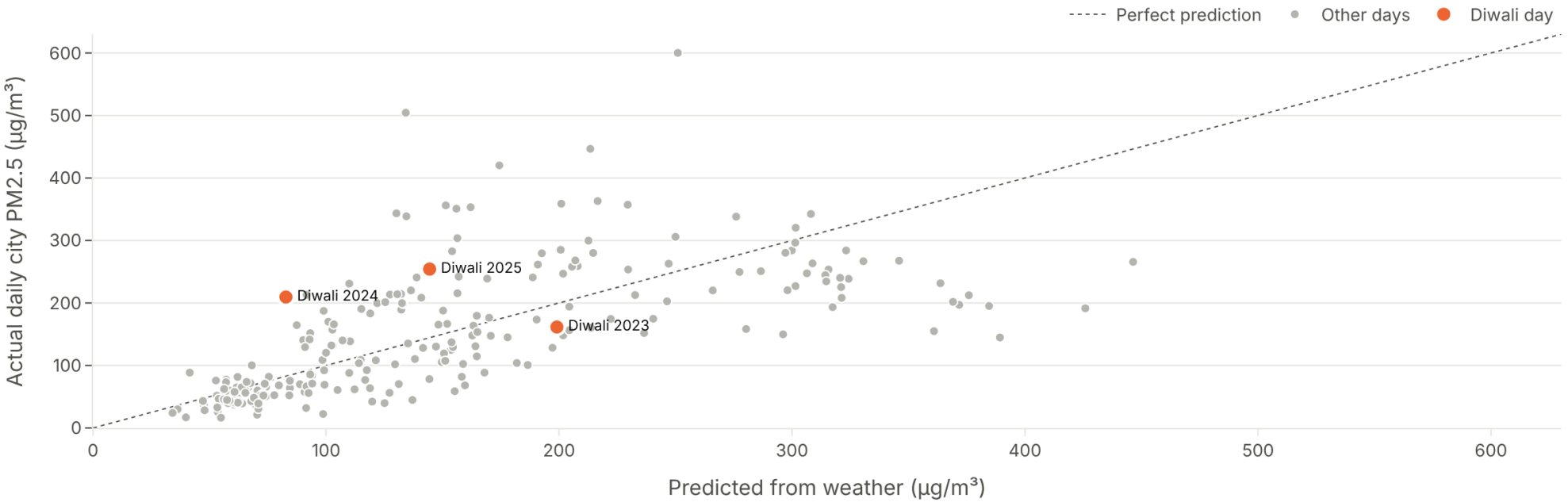
West and north-west winds blow 60% of the time; west winds bring the dirtiest air



By wind direction (October–November). West and north-west winds each blew about **30% of the time**. West winds brought the dirtiest air (**206 µg/m³**); north-west winds, from Punjab's direction, the cleanest (**157**), and they were also the strongest (8.0 vs 6.5 km/h).

FIGURE 17 · WEATHER

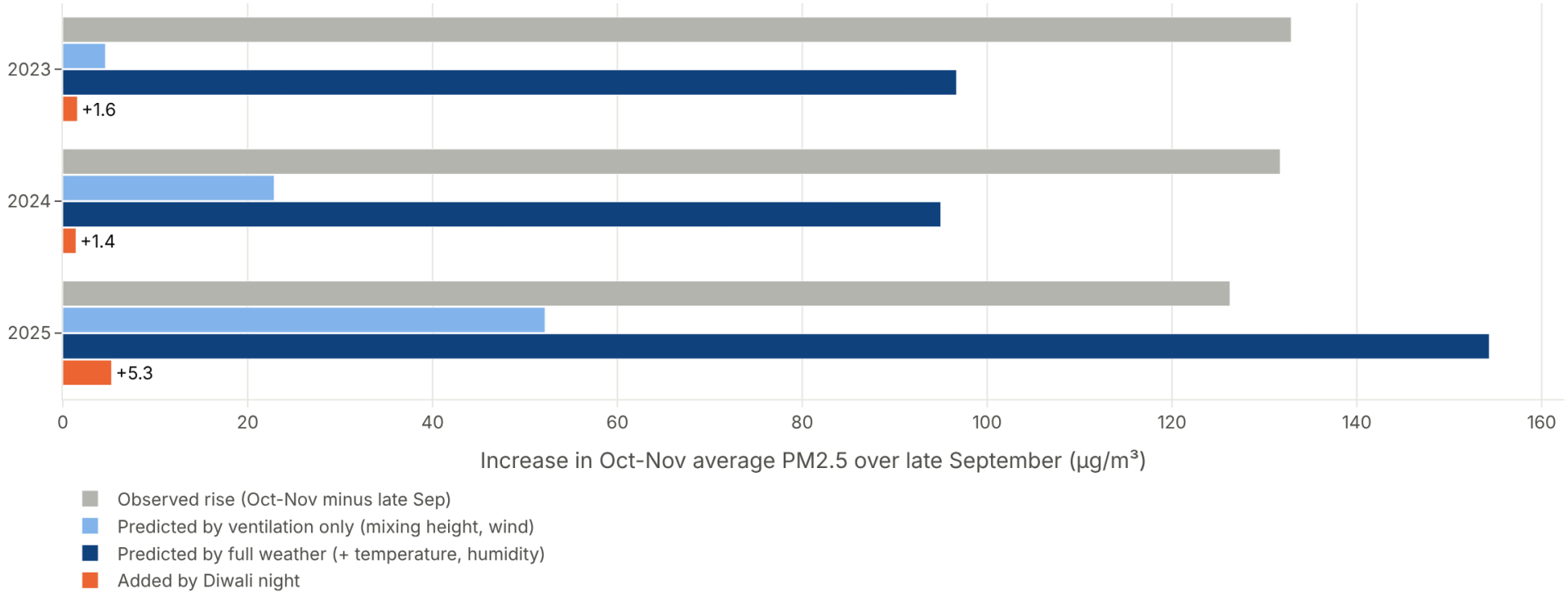
The full-weather model follows the season (R^2 = 0.63) but misses the sharpest spikes



What the full-weather model predicts against what happened, day by day. It captures the seasonal level (**R^2 = 0.63**) but misses the sharpest episodes, such as 18 November 2024 (600 actual vs 251 predicted).

FIGURE 18 · WEATHER

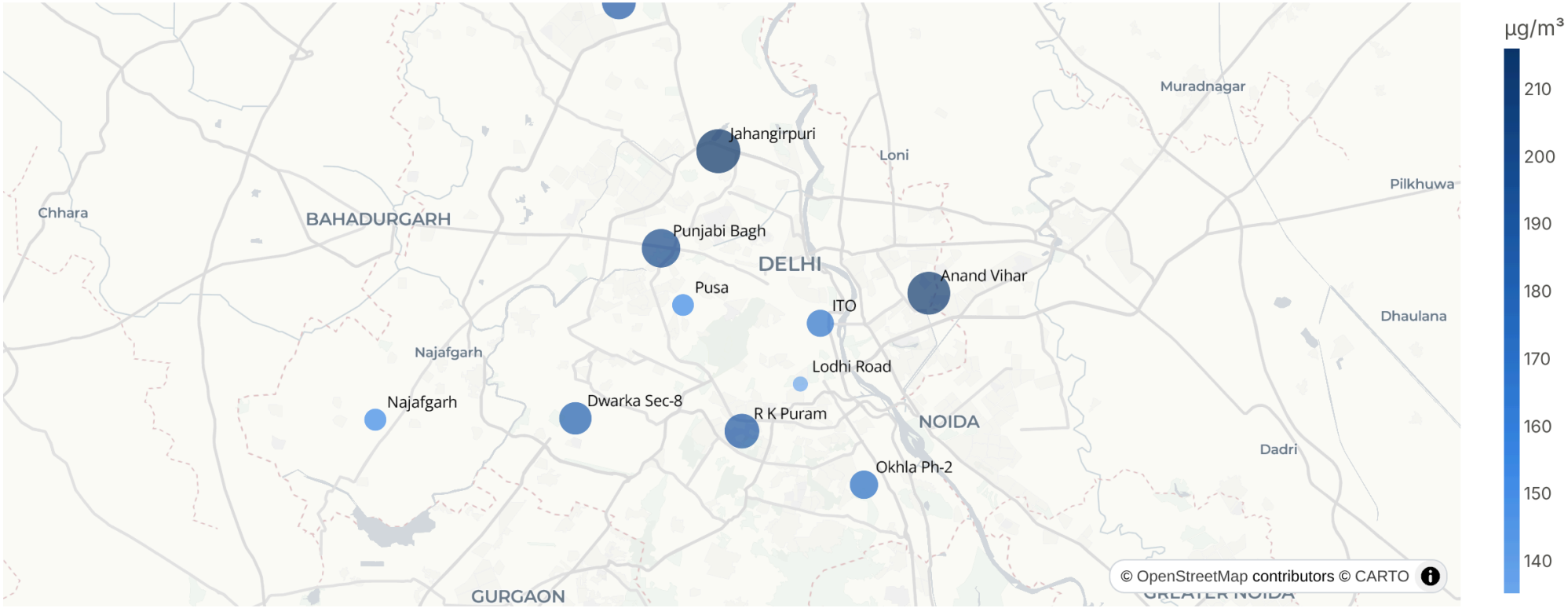
The season, not Diwali night, drives the October-November rise



The answer to the question. The seasonal rise was **+133, +132 and +126 µg/m³**. Ventilation alone (mixing height, wind) predicts **3–41%** of it; adding temperature and humidity, **72–122%**. Diwali night adds just **+1.4 to +5.3 µg/m³** to the October–November average: **1.1–4.2%** of the rise.

FIGURE 19 · STATIONS

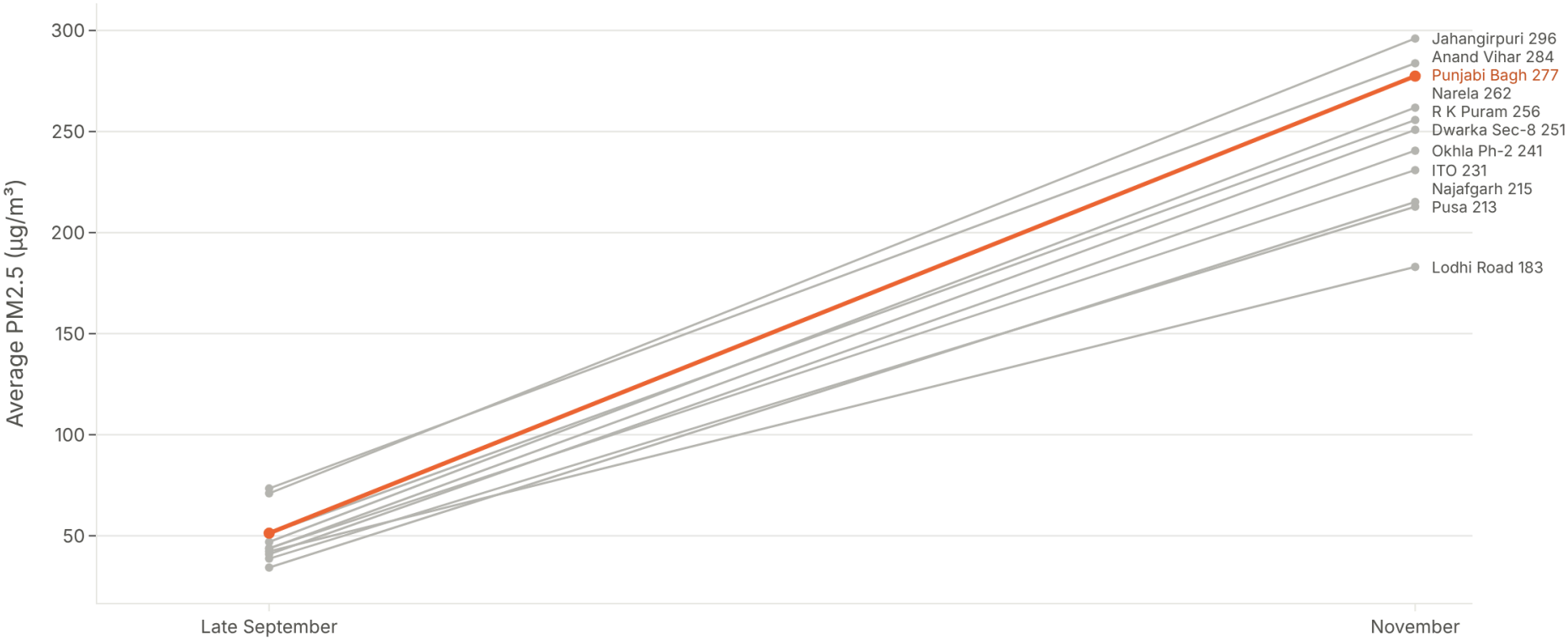
Jahangirpuri (north) and Anand Vihar (east) are the most polluted stations; Lodhi Road the least



October–November averages. **Jahangirpuri (216 µg/m³)** in the north and **Anand Vihar (213)** in the east are the most polluted; **Lodhi Road (135)** in central Delhi is the cleanest, and still more than twice India's limit.

FIGURE 20 · STATIONS

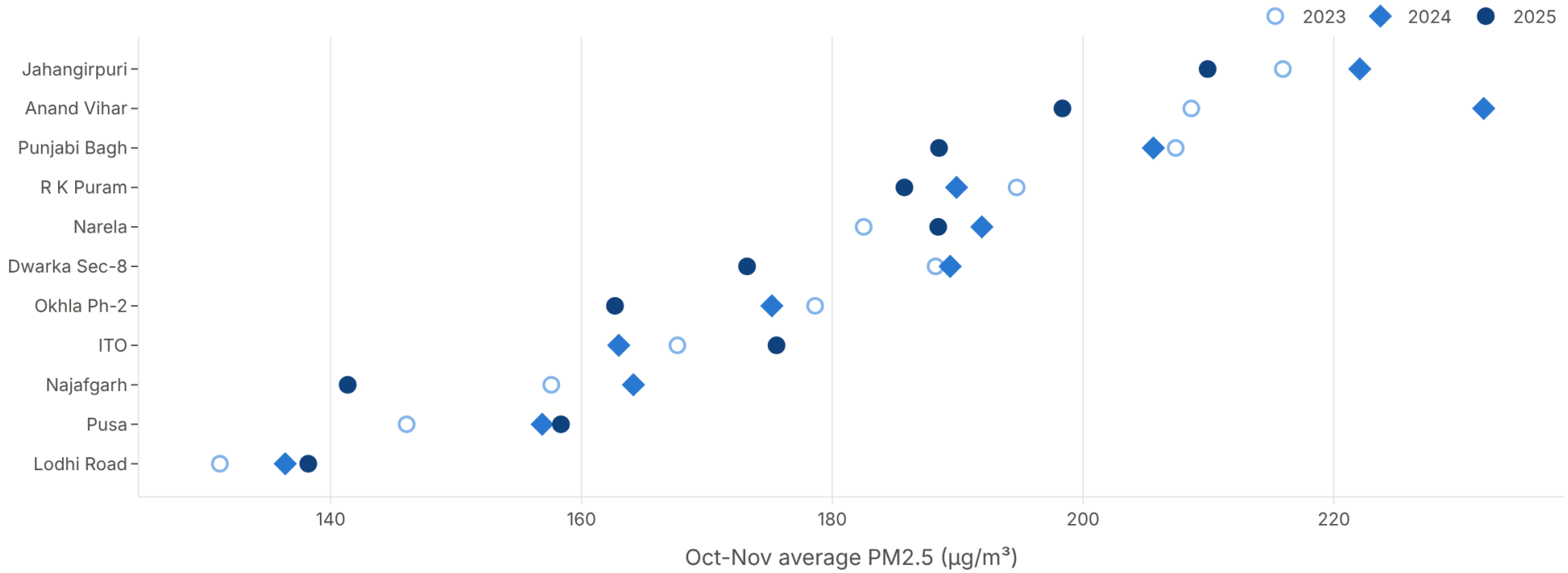
Every station climbs steeply; Punjabi Bagh rises the most (+226 µg/m³)



Each station from late September to November (averages over three years). Late-September levels range from 34 to 73 µg/m³; November from 183 to 296. **Punjabi Bagh** rises the most (+226), and Lodhi Road the least (+141).

FIGURE 21 · STATIONS

The ranking is stable: the same top three and the same cleanest station every year



The ranking hardly moves between years (rank correlation **0.92–0.95**). Jahangirpuri, Anand Vihar and Punjabi Bagh are the top three every year, and Lodhi Road the cleanest. That suggests local sources that stay in the same places year after year.